UNIX/LINUX Operating Systems

Dr. Y. Yang
Updated by Dr. M. Kim

CS Department, PVAMU

Frequently Used UNIX/LINUX Commands

passwd

Change your password. First, it asks for your old password, then prompts for new password.

/etc/passwd

All legal users' information, including name, password, group, home directory, etc., is stored in this file. This file is an editable file in text format. If you are the administrator, you can add/delete users by changing this file. If somebody forgets his/her password, you can modify his/her record in this file and remove his/her password.

cal

This command will print a calendar for a specified month and/or year.
To show this month's calendar, enter:

```
cal
```

To show a twelve-month calendar for 2018, enter:

```
cal 2018
```

To show a calendar for just the month of June, 1970, enter:

```
cal 6 1970
```

cat

This command outputs the contents of a text file. You can use it to read brief files or to concatenate files together.
To append `file1` onto the end of `file2`, enter:

```
cat file1 >> file2
```

To view the contents of a file named `myfile`, enter:

```
cat myfile
```

Because `cat` displays text without pausing, its output may quickly scroll off your screen. Use the `less` command (described below) or an editor for reading longer text files.

cd

This command changes your current directory location. By default, your Unix login session begins in your home directory.
To switch to a subdirectory (of the current directory) named `myfiles`, enter:

```
cd myfiles
```

To switch to a directory named `/home/dvader/empire_docs`, enter:

```
cd /home/dvader/empire_docs
```

To move to the parent directory of the current directory, enter:

```
cd ..
```

To move to the root directory, enter:

```
cd /
```

To return to your home directory, enter:

```
cd
```

chmod

This command changes the permission information associated with a file. Every file (including directories, which Unix treats as files) on a Unix system is stored with records indicating who has permission to read, write, or execute the file, abbreviated as r, w, and x. These permissions are broken down for three categories of user: first, the owner of the file; second, a group with which both the user and the file may be associated; and third, all other users. These categories are abbreviated as u for owner (or user), g for group, and o for other.

To allow yourself to execute a file that you own named `myfile`, enter:

```
chmod u+x myfile
```

To allow anyone who has access to the directory in which `myfile` is stored to read or execute `myfile`, enter:

```
chmod o+rx myfile
```

You can view the permission settings of a file using the `ls` command, described below.

Note: Be careful with the `chmod` command! If you tamper with the directory permissions of your home directory, for example, you could lock yourself out or allow others unrestricted access to your account and its contents.

cp

This command copies a file, preserving the original and creating an identical copy. If you already have a file with the new name, `cp` will overwrite and destroy the duplicate. For this reason, it's safest to always add `-i` after the `cp` command, to force the system to ask for your approval before it destroys any files. The general syntax for `cp` is:

```
cp -i oldfile newfile
```

To copy a file named `meeting1` in the directory `/home/dvader/notes` to your current directory, enter:

```
cp -i /home/dvader/notes/meeting1 .
```

The `.` (period) indicates the current directory as destination, and the `-i` ensures that if there is another file named `meeting1` in the current directory, you will not overwrite it by accident.

To copy a file named `oldfile` in the current directory to the new name `newfile` in the `mystuff` subdirectory of your home directory, enter:

```
cp -i oldfile ~/mystuff/newfile
```

The `~` character (tilde) is interpreted as the path of your home directory.

Note: You must have permission to read a file in order to copy it.

date

The `date` command displays the current day, date, time, and year.

To see this information, enter:

```
date
```

du

This command reports disk usage (i.e., the amount of space taken up by a group of files). `du` descends all subdirectories from the directory in which you enter the command, reporting the size of their contents, and finally reporting a total size for all the files it finds.
To find out how much disk space your files take up, switch to your home directory with the `cd` command, and enter:

```
du
```

The numbers reported are the sizes of the files; on different systems, these sizes will be in units of either 512 byte blocks or kilobytes. To learn which is the case, use the `man` command, described below. On most systems, `du -k` will give sizes in kilobytes.

find

The `find` command lists all of the files within a directory and its subdirectories that match a set of conditions. This command is most commonly used to find all of the files that have a certain name.
To find all of the files named `myfile.txt` in your current directory and all of its subdirectories, enter:

```
find . -name myfile.txt -print
```

To look in your current directory and its subdirectories for all of the files that end in the extension `.txt` , enter:

```
find . -name "*.txt" -print
```

In these examples, the `.` (period) represents your current directory. It can be replaced by the full pathname of another directory to search. For instance, to search for files named `myfile.txt` in the directory `/home/user/myusername` and its subdirectories, enter:

```
find /home/user/myusername/ -name myfile.txt -print
```

On some systems, omitting the final `/` (slash) after the directory name can cause `find` to fail to return any results.
As a shortcut for searching in your home directory, enter:

```
find "$HOME/" -name myfile.txt -print
```

finger

The `finger` command reports information about an account holder, including name, office, phone number (if any), last login, and other optional information.
To find out more about user `dvader`, enter:

```
finger dvader
```

If the computer on which `dvader` has his account is different from the system you are using, you may need to specify the other computer's name, for example:

```
finger dvader@deathstar
```

If `dvader` has his account on a computer at a different campus, university, or organization, you will need to specify his complete address, for example:

```
finger dvader@deathstar.empire.gov
```

jobs

This command reports any programs that you suspended and still have running or waiting in the background (if you had pressed `Ctrl-z` to suspend an editing session, for example). For a list of suspended jobs, enter:

```
jobs
```

Each job will be listed with a number; to resume a job, enter `%` (percent sign) followed by the number of the job. To restart job number two, for example, enter:

```
%2
```
This command is only available in the csh, bash, tcsh, and ksh shells.

kill

Use this command as a last resort to destroy any jobs or programs that you suspended and are unable to restart. Use the jobs command to see a list of suspended jobs. To kill suspended job number three, for example, enter:
```
kill %3
```
Now check the jobs command again. If the job has not been cancelled, harsher measures may be necessary. Enter:
```
kill -9 %3
```

less

This utility displays the contents of a text file one screen at a time, waiting for you to press the Spacebar between screens. This lets you read text without it scrolling quickly off your screen. The less utility is a more flexible and powerful version of more. For example, unlike more, less allows you to scroll backwards by pressing b .
To read the contents of a file named textfile in the current directory, enter:
```
less textfile
```
The less utility is often used for reading the output of other commands. For example, to read the output of the ls command one screen at a time, enter:
```
ls -la | less
```
To exit less after viewing the file, press q .
Note: Don't use less with executables (binary files), such as output files produced by compilers. Doing so will display garbage and may lock up your terminal.

lpr and lp

These commands print a file on a printer connected to the computer network. The lpr command is used on BSD systems, and the lp command is used in System V.
To print a file named myfile on a printer named lp1 with lpr, enter:
```
lpr -Plp1 myfile
```
To print the same file to the same printer with lp, enter:
```
lp -dlp1 myfile
```
Note: Do not print to a printer whose name or location is unfamiliar to you; you may annoy its owners.

ls

This command will list the files stored in a directory. To see a brief, multi-column list of the files in the current directory, enter:
```
ls
```
To also see "dot" files (configuration files that begin with a period, such as .login), enter:
```
ls -a
```
To see the file permissions, owners, and sizes of all files, enter:
```
ls -la
```
If the listing is long and scrolls off your screen before you can read it, combine ls with the less utility, for example:
```
ls -la | less
```

man

This command displays the manual page for a particular command. If you are unsure how to use a command or want to find out all its options, you might want to try using `man` to view the manual page.
For example, to learn more about the `ls` command, enter:

```
man ls
```

To learn more about `man`, enter:

```
man man
```

If you aren't sure of the exact command name, you can use `man` with the `-k` option to help you find the command you need. To see one line summaries of each reference page that contains the keyword you specify, enter:

```
man -k keyword
```

Replace `keyword` in the above example with the keyword which you want to reference.

mesg

This command toggles permission for others to send messages to your terminal. If someone were bothering you with requests to use `talk`, for example, you could disable the messages and ignore the person by entering:

```
mesg n
```

If you had a change of heart, you could again enable messages by entering:

```
mesg y
```

Note: In most applications, you can restore screen contents that were disrupted by incoming messages by pressing `Ctrl-l` (the letter l).

mkdir

This command will make a new subdirectory.
To create a subdirectory named `mystuff` in the current directory, enter:

```
mkdir mystuff
```

To create a subdirectory named `morestuff` in the existing directory named `/tmp`, enter:

```
mkdir /tmp/morestuff
```

Note: To make a subdirectory in a particular directory, you must have permission to write to that directory.

more

This utility displays the contents of a text file, one screen at a time, waiting for you to press the Spacebar between screens. This lets you read text without it scrolling quickly off your screen. `less`, described above, is a more powerful version of `more`.
To read the contents of a file named `textfile` in the current directory, enter:

```
more textfile
```

The `more` utility is often used for reading the output of other commands. For example, to read the output of the `ls` command one screen at a time, enter:

```
ls -la | more
```

Note: Don't use `more` with executables (binary files), such as output files produced by compilers. Doing so will display garbage and may lock up your terminal.

mv

This command will move a file. You can use `mv` not only to change the directory location of a file, but also to rename files. Unlike the `cp` command, `mv` will not preserve the original file.
Note: As with the `cp` command, you should always use `-i` to make sure you don't overwrite an existing file.
To rename a file named `oldname` in the current directory to the new name `newname`, enter:

```
mv -i oldname newname
```

To move a file named `hw1` from a subdirectory named `newhw` to another subdirectory named `oldhw` (both subdirectories of the current directory), enter:

```
mv -i newhw/hw1 oldhw
```

If, in this last operation, you also wanted to give the file a new name, such as `firsthw`, you would enter:

```
mv -i newhw/hw1 oldhw/firsthw
```

pwd

This command reports the current directory path. Enter the command by itself:

```
pwd
```

rm

This command will remove (destroy) a file. You should enter this command with the `-i` option, so that you'll be asked to confirm each file deletion. To remove a file named `junk`, enter:

```
rm -i junk
```

Note: Using `rm` will remove a file permanently, so be sure you really want to get rid of the file before you use `rm`.

rmdir

This command will remove a subdirectory. To remove a subdirectory named `oldstuff`, enter:

```
rmdir oldstuff
```

Note: The directory you specify for removal must be empty. To clean it out, switch to the directory and use the `ls` and `rm` commands to inspect and delete files.

set

This command displays or changes various settings and options associated with your Unix session. To see the status of all settings, enter the command without options:

```
set
```

If the output scrolls off your screen, combine `set` with `less`:

```
set | less
```

The syntax used for changing settings is different for the various kinds of Unix shells; see the `man` entries for `set` and the references listed at the end of this document for more information.

w

This command lists all users currently logged into the computer, and includes brief information about what they are doing. To see the listing, just enter `w` .

who

This command will list all users logged into the computer, and the dialup connection or host computer from which they are connected. To see the listing, just enter `who` .

ps -x

See what processes (programs) are running.

ps -ax

See all processes running on the box.

quota

Check how many available space you have, and what you have used so far.

top

Take a look at the top resource eating processes on the system.

tar

Compresses or decompress files.
tar –cvf filename.tar files //compress
tar –xvf filename.tar //decompress

gzip

Compresses files. The compressed file gets the extension .gz and replaces the original file.
gzip -5 *filename*

gunzip

Decompresses the files. The decompressed file looses the ``compression extension", ie. .gz, and replaces the compressed file.
gunzip *filename*

wc

Shows the number of lines, words and characters in the file(s) filename(s), both as individuals and as a total of all files.
wc filename(s)

vi

This command starts the vi text editor. To edit a file named `myfile` in the current directory, enter:

```
vi myfile
```

The vi editor works fairly differently from other text editors. The very least you need to know to start using vi is that in order to enter text, you need to switch the program from command mode to insert mode by pressing `i` . To navigate around the document with the cursor keys, you must switch back to command mode by pressing `Esc`. To execute any of the following commands, you must switch from command mode to ex mode by pressing `:` (the colon key): Enter `w` to save; `wq` to save and quit; `q!` to quit without saving.

pico

Pico is probably the most popular and easiest to use text editor. Here the basic commands that you will use the most. Pico comes with a nice help, so refer to that when you need more extensive help then the basics I give you here :o)
pico filename Starts up pico with the filename loaded. If the filename didn't exist, you create a new file with that name.
ctrl-v scroll one page down
ctrl-y scroll one page up
ctrl-c Shows you the line number the cursor is on, handy when your getting errors *ctrl-w* With this you can search for words etc in your file.
ctrl-x Quit pico, if file is changed, it will ask if you want to save it.
ctrl-g Pico's help pages.

telnet

ftp

cd, lcd, get, put, ls, …

ping

useradd -m hkim -d /home/hkim -s /bin/bash -- only root can do this

Adding user hkim, create a folder hkim under /home and when user login in the current location will be /home/hkim

passwd hkim --- as a root, you can reset user's (e.g., hkim) password. Type twice new password

sudo rm -r hkim --- execute as a root. Delete everything (recursive hkim and below it)

use g++ (if you use gcc – gnu C compiler, it does not work with cout statement. gcc test.c – C program is okay with gcc)

g++ test.cc -- compile c++ code, default output file name is a.out
./a.out -- execute

cat /etc/*release
current OS version checking

ps –ef |more
display all process currently running

grep -rl ftruncate /usr/include
search for the all files under include which contain 'ftruncate' in their files.